

Numerical Stiffness

- ❑ **Definition:** Consider a system which can be represented by a system of ODEs (generally as a nonlinear function).

The system of ODEs is called **numerically stiff** (or simply **stiff**) if it contains rapidly decaying (**fast**) dynamics alongside slowly varying (**slow**) dynamics.

$$\begin{cases} \frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}, t) \\ \mathbf{x}(t_0) = \mathbf{x}_0 \in \mathcal{R}^m \end{cases}$$

- ❖ Even though the fast dynamics die out quickly, they constrain the numerical solver to take **very small step sizes** for stability, much smaller than what would be needed to solve for the slow dynamics.



- Simultaneous presence of widely separated time scales in the system



- Explicit solvers become **unstable** unless the step size is very small (so it's about numerical stability)

- ❖ **Example:**

$$\begin{cases} \frac{d}{dt} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -1000 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ \begin{bmatrix} x_1(0) & x_2(0) \end{bmatrix}^T = \begin{bmatrix} 10 & 10 \end{bmatrix}^T \end{cases}$$

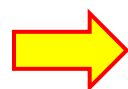
Recall:

$$\frac{dx}{dt} = -ax \quad \Rightarrow \quad x = Ae^{-at}$$

Numerical Stiffness

❖ Consider an LTI system:

$$\begin{cases} \frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x} \\ \mathbf{x}(t_0) \neq \mathbf{x}_0 \end{cases}$$



Solution: $\mathbf{x} = e^{\mathbf{A}t} \mathbf{x}_0$

Taylor's Series: $e^{\mathbf{A}t} = \sum_{n=0}^{\infty} \frac{(\mathbf{A}t)^n}{n!}$

➤ Suppose the matrix $\mathbf{A}$ can be expressed as $\mathbf{A} = \mathbf{S}\mathbf{\Lambda}\mathbf{S}^{-1}$ such that:

$$\mathbf{\Lambda} = \begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \lambda_m \end{bmatrix}$$

$\lambda_i \in \mathbb{C}$

$\lambda_1, \lambda_2, \dots, \lambda_m$ are called the eigenvalues of the matrix $\mathbf{A}$

$$\det(\lambda_i \mathbf{I} - \mathbf{A}) = 0$$

❖ If a square matrix $\mathbf{A}$ acts as a linear transformation on a vector $\mathbf{x}$, in general, multiplying the matrix changes both the length and the direction of the vector. However, some special vectors keep their direction the same (they only get stretched, compressed, or flipped). These special vectors are eigenvectors, and the factors by which they are stretched are eigenvalues.

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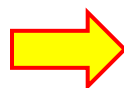
❖ $\mathbf{v}_i \in \mathbb{C}^m$ is an eigenvector of $\mathbf{A}$ if $(\lambda_i \mathbf{I} - \mathbf{A})\mathbf{v}_i = 0$ or $\mathbf{A}\mathbf{v}_i = \lambda_i \mathbf{v}_i$

✓ It can also be proven that $\mathbf{S} = [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_m]$

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$$\begin{cases} \frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x} \\ \mathbf{x}(t_0) \neq \mathbf{x}_0 \end{cases}$$



Solution:

$$\begin{aligned} \mathbf{x} &= e^{\mathbf{A}t} \mathbf{x}_0 \\ \mathbf{A} &= \mathbf{S}\mathbf{\Lambda}\mathbf{S}^{-1} \quad \mathbf{\Lambda} = \begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \lambda_m \end{bmatrix} \end{aligned}$$

$$\Rightarrow \mathbf{x}(t) = e^{\mathbf{A}t} \mathbf{x}_0 = e^{\mathbf{S}\mathbf{\Lambda}\mathbf{S}^{-1}t} \mathbf{x}_0 = \mathbf{S} e^{\mathbf{\Lambda}t} \mathbf{S}^{-1} \mathbf{x}_0$$

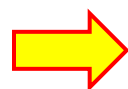
$$e^{\mathbf{\Lambda}t} =$$

$$\begin{cases} \checkmark e^{\lambda_i t} & \text{defines the internal modes (increasing, decreasing)} \\ \checkmark \lambda_i & \text{determines how the modes in } \mathbf{x}(t) \text{ change (fast, slow)} \end{cases}$$

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Solution:

$$\mathbf{x} = \mathbf{S}e^{\mathbf{A}t}\mathbf{S}^{-1}\mathbf{x}_0$$

$$e^{\mathbf{A}t} = \begin{bmatrix} e^{\lambda_1 t} & & \\ & e^{\lambda_2 t} & \\ & & e^{\lambda_m t} \end{bmatrix}$$

- $$\begin{cases} \checkmark e^{\lambda_i t} & \text{defines the internal modes (increasing, decreasing)} \\ \checkmark \lambda_i & \text{determines how the modes in } \mathbf{x}(t) \text{ change (fast, slow)} \end{cases}$$

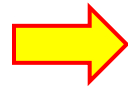
❖ **Stability of the LTI System:**

- If for any eigenvalue $\text{Re}(\lambda_i) > 0$, the term $e^{\lambda_i t}$ grows exponentially in time and the (actual) system is unstable
- ✓ Let's assume that we are simulating a stable system, i.e., all eigenvalues have $\text{Re}(\lambda_i) < 0$

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Solution:

$$\mathbf{x} = \mathbf{S}e^{\Lambda t}\mathbf{S}^{-1}\mathbf{x}_0$$

$$e^{\Lambda t} = \begin{bmatrix} e^{\lambda_1 t} & & \\ & e^{\lambda_2 t} & \\ & & e^{\lambda_m t} \end{bmatrix}$$

$$\begin{cases} \checkmark e^{\lambda_i t} \text{ defines the internal modes (increasing, decreasing)} \\ \checkmark \lambda_i \text{ determines how the modes in } \mathbf{x}(t) \text{ change (fast, slow)} \end{cases}$$

❖ Stiffness Ratio: indicates how much different fast and slow dynamics are!

$$|Re(\lambda_{\min})| \ll |Re(\lambda_{\max})| \quad \text{or} \quad S = \frac{|Re(\lambda_{\max})|}{|Re(\lambda_{\min})|} = 10^3 \dots 10^{10}$$

➤ Stiff systems are harder to solve due to numerical instability!

Numerical Stability

To be continued...